SMART WATER QUALITY MONITORING SYSTEM

An IoT-Based Smart Water Purification Solution

Presented By:

NOVA MINDS

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Technology:

ESP32, IoT, Sensors

THE PROBLEM

KEY POINTS



Water quality is not always **monitored regularly.**



Traditional purification systems may not provide **real-time** water-quality information.



Users may not know when **water quality changes.**



Manual monitoring requires additional **time and effort.**



OUR GOAL: To develop an affordable smart system for monitoring water quality and controlling the purification process.

OUR SOLUTION

We propose an **IoT-based** smart water purification system that monitors water quality using sensors, sends data to **Firebase**, and displays the readings on a mobile application.

KEY FEATURES:



TDS and temperature measurement.



Automatic purification control.



Data stored in Firebase and shown on mobile application.



Real-time data view on mobile app.



Alerts when readings exceed configured limits.



Before-and-after purification comparison.

HARDWARE REQUIREMENTS

Components used to build our smart water quality monitoring and purification system.

COMPONENTS USED:



ESP32 Microcontroller



Turbidity Sensor



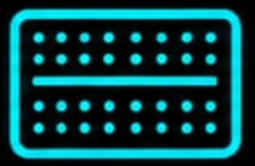
TDS Sensor



Temperature Sensor



Relay Module



Breadboard

PURPOSE: These components work together to monitor water quality and control the purification process.

WORKING PRINCIPLE



WATER INPUT → TDS, TURBIDITY & TEMPERATURE SENSORS → ESP32 DATA PROCESSING → FIREBASE CLOUD → MOBILE APPLICATION

SYSTEM WORKFLOW



Sensors measure TDS, turbidity and temperature of the water.



ESP32 collects the readings, processes the data and sends it to Firebase through Wi-Fi.



Firebase stores the readings and displays the data on our mobile application.

Real-time monitoring enables users to view water quality readings anytime.

FUTURE SCOPE



AI-based water-quality trend analysis.



Cloud-based storage of sensor readings.



Automatic filter replacement **reminders.**



Water **consumption** monitoring.



Improved detection of additional **contaminants.**



Integration with **smart homes** and **community water systems.**

CONCLUSION

The **Smart Water Purification and Quality Monitoring System** combines sensors, embedded systems, and IoT technology to improve water-quality monitoring.



It provides **real-time measurements**, remote monitoring, and automated control of the purification process.





Our Vision:
To make water-quality monitoring **smarter**, **more accessible**, and **more efficient.**

THANK YOU!